pfi

Python for Fixed Income

COURSE 5B · TIME SERIES MODELS

Autocorrelation: ACF and PACF

WEEK 1 · DAY 4

Created by **Jeff Lenamon**

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Today: how much does a series remember

Scope

Development years only: 2 Jan 2015 to 30 Dec 2022, 2,001 dates. Nothing is fitted or scored today: we measure, and read what we measured.

Six things

1. Lag scatter plots
2. The autocorrelation by hand
3. A bar against its band
4. The partial autocorrelation
5. The identification grid
6. The change, its size, and Ljung-Box

The change has almost no memory. Its square has a lot.

Names for a bar

An autocorrelation is a correlation of a series with itself: a bar, a lag, a band. A model's output, when we get to models:

Allowed names

fitted change in bp

the model's one-step value

the last value carried forward

Never called

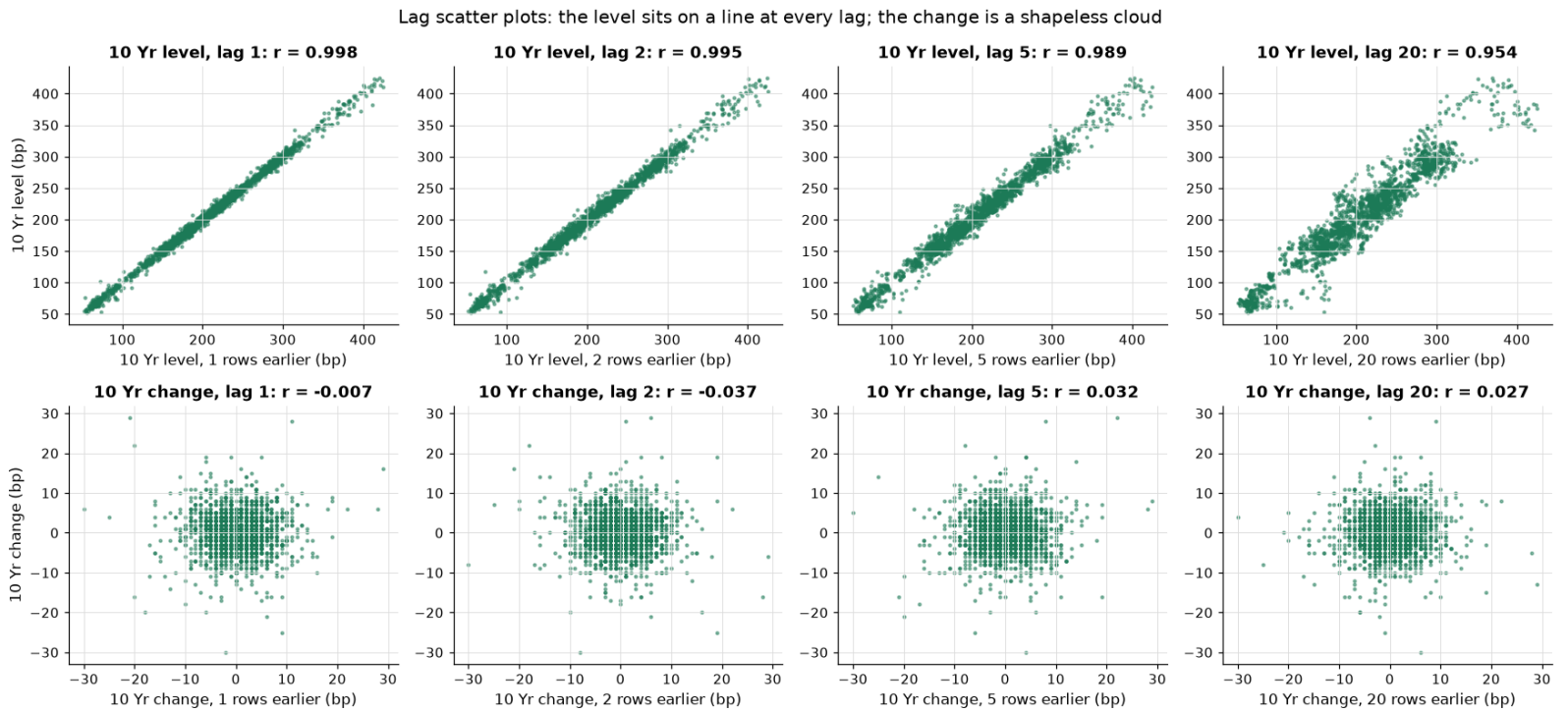
a forecast or a prediction

a projection or an outlook

a signal, a call, a view on rates

Nothing today says what a yield will do.

Lag scatter plots



The level lies on a line at every lag; the change is a shapeless cloud. The correlation of a series with itself at a lag is the autocorrelation at that lag.

How the autocorrelation works

The problem	How strongly is a value tied to the value k rows before it?
Deviations	Take the series' average away from every value
The ratio	Add the products of each deviation and the one k rows earlier; divide by the sum of squared deviations
You control	The lag; at lag 0 the answer is exactly 1
What comes out	A number from -1 to 1 per lag; the list over lags 1, 2, 3 is the ACF

Three ways to the same number

Lag	By hand	statsmodels	pandas
1	-0.00747	-0.00747	-0.00748
2	-0.03725	-0.03725	-0.03731
3	-0.01716	-0.01716	-0.01719
4	-0.00076	-0.00076	-0.00076
5	0.03164	0.03164	0.03176

acf_by_hand

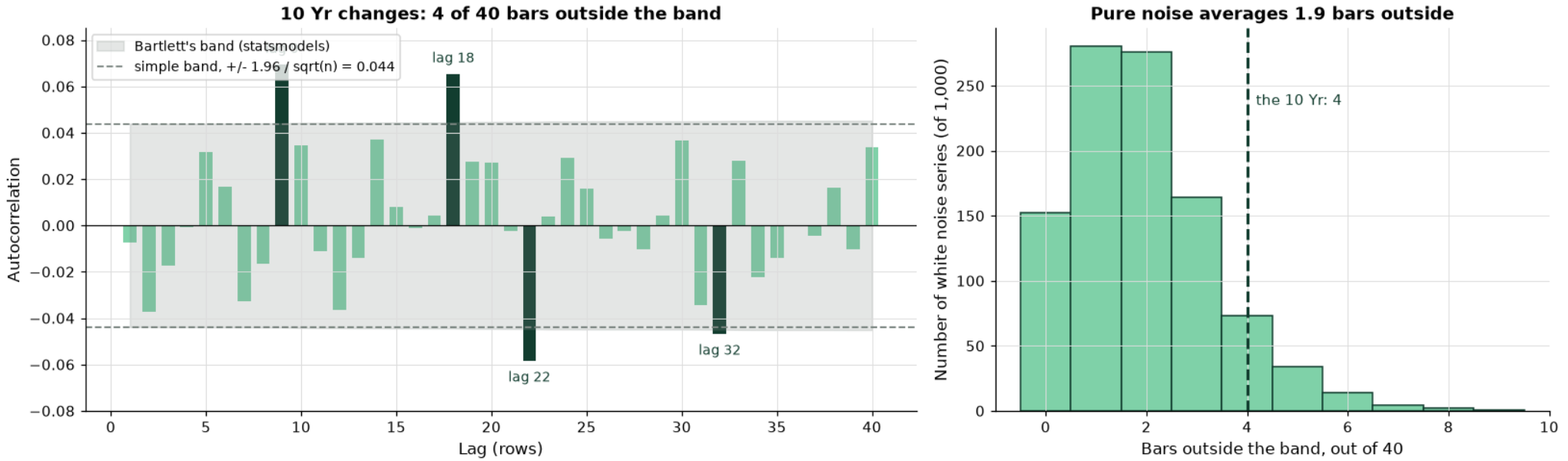
By hand and statsmodels agree to 1e-12 at every lag. Pandas takes the plain correlation of the two offset stretches, so it differs a little. 21 tests pass.

Every bar is close to zero: yesterday's change explains almost none of today's.

How the confidence band works

The problem	A sample autocorrelation is never exactly zero, even for a series with no memory
The simple band	About 95 percent of bars fall inside +/- 1.96 / sqrt(n) when there is no autocorrelation
Bartlett's band	statsmodels' wider band, growing with the lag
Reading a bar	Inside the band: indistinguishable from zero. Outside: a candidate for memory, not proof
Chance	One bar in twenty is outside by chance alone

A bar against its band, and chance

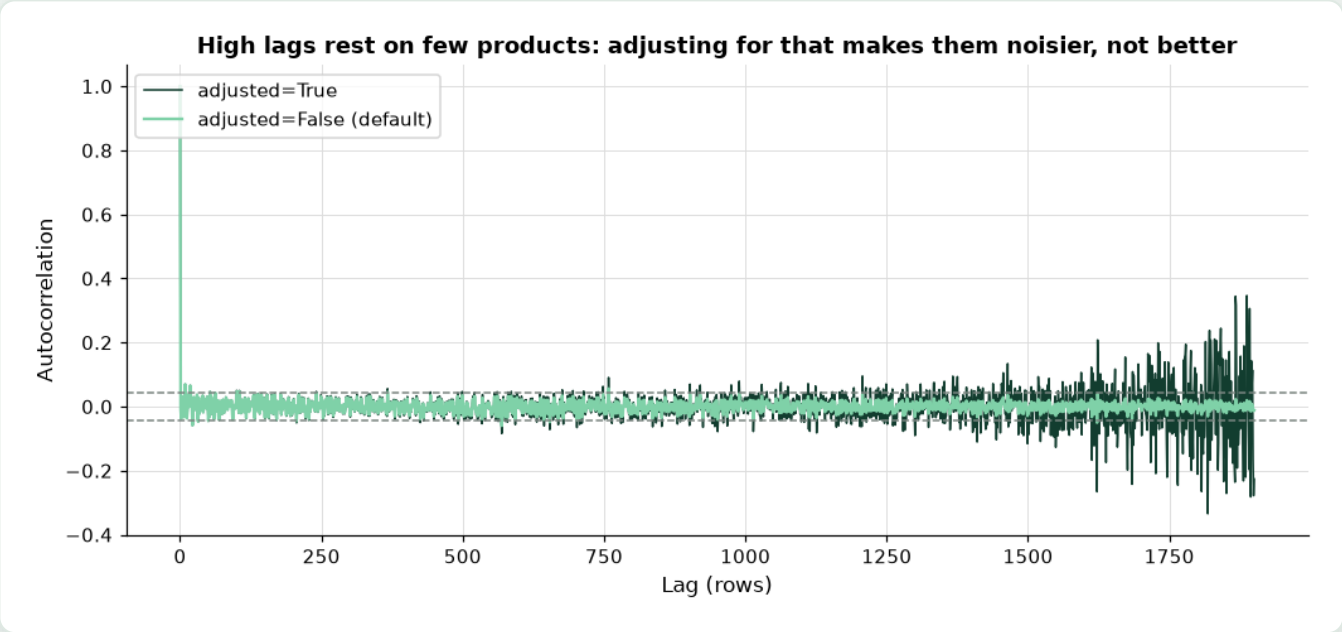


Simple band 0.0438; Bartlett's 0.0438 at lag 1, 0.0451 at lag 40. 4 of 40 bars outside, at lags 9, 18, 22, 32. Noise averages 1.92 outside; 84.8% of noise series have at least one, 12.8% have four or more.

Read the shape, not every bar. A pattern is a run at lags 1, 2, 3, or a regular beat.

Every setting of acf, and adjusted

Setting	Default	Tune first
nlags	None	yes
adjusted	False	yes
qstat	False	no
fft	True	no
alpha	None	yes
bartlett_confint	True	no
missing	'none'	no



From lag 1,800 to 1,900 the largest bar is 0.030 unadjusted and 0.347 adjusted: a high lag rests on few products.

The default, adjusted=False, keeps high lags small: the course uses it.

How the partial autocorrelation works

The problem	The lag-2 autocorrelation mixes the direct tie with the tie that runs through yesterday
The definition	Regress the value on its first k lags: the PACF at lag k is the last coefficient
Lag 1	The PACF equals the ACF
You control	The lag and the method (Yule-Walker, regressions, Levinson-Durbin, Burg)
What comes out	One bar per lag, read against the same band as the ACF

The PACF is a regression

Lag	ywadjusted	ywm	ols	burg
1	-0.00748	-0.00747	-0.00748	-0.00748
2	-0.03734	-0.03731	-0.03735	-0.03737
3	-0.01778	-0.01776	-0.01780	-0.01781
4	-0.00244	-0.00243	-0.00241	-0.00241

Lag 2 as a regression

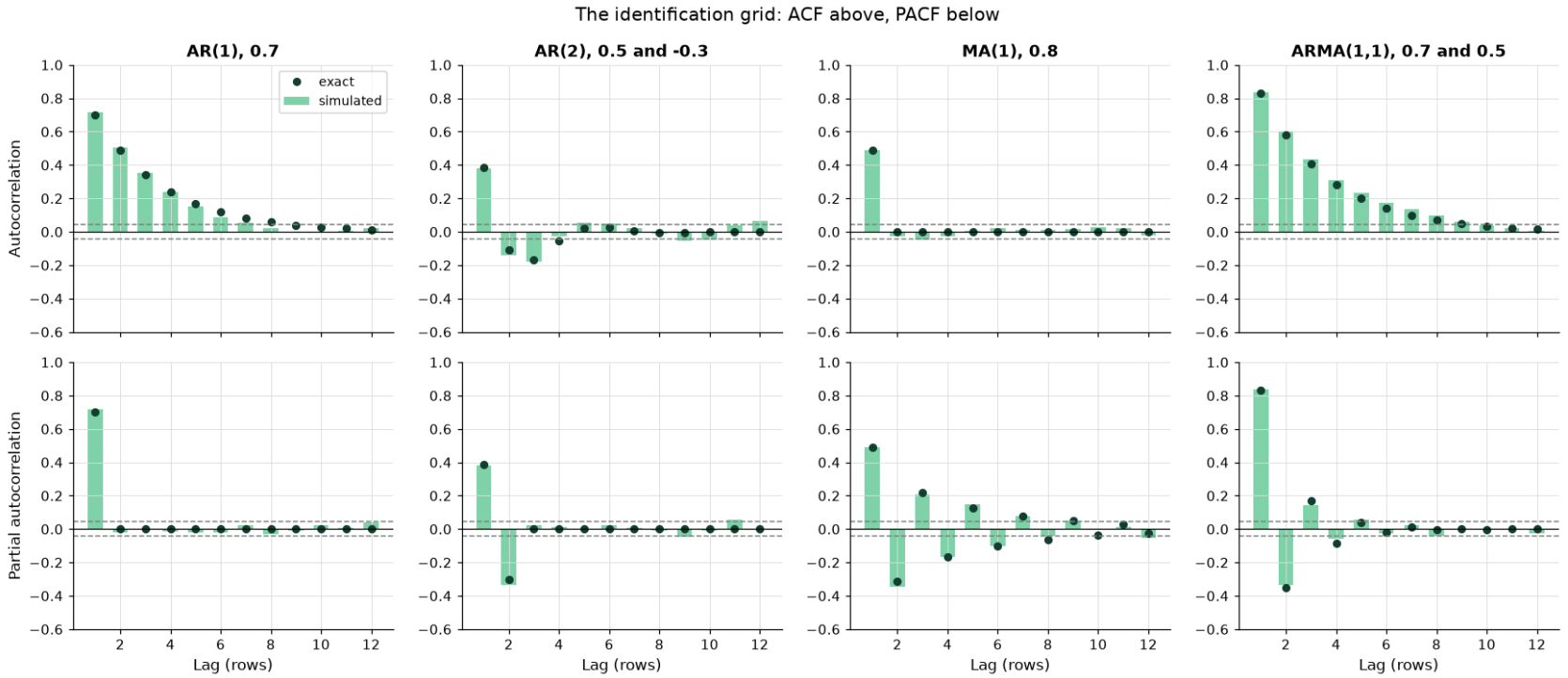
The last coefficient of the regression is -0.037348; the ols method gives -0.037348: equal to 1e-12.

On a long series the method hardly matters. Two settings matter: nlags and method.

How identification works

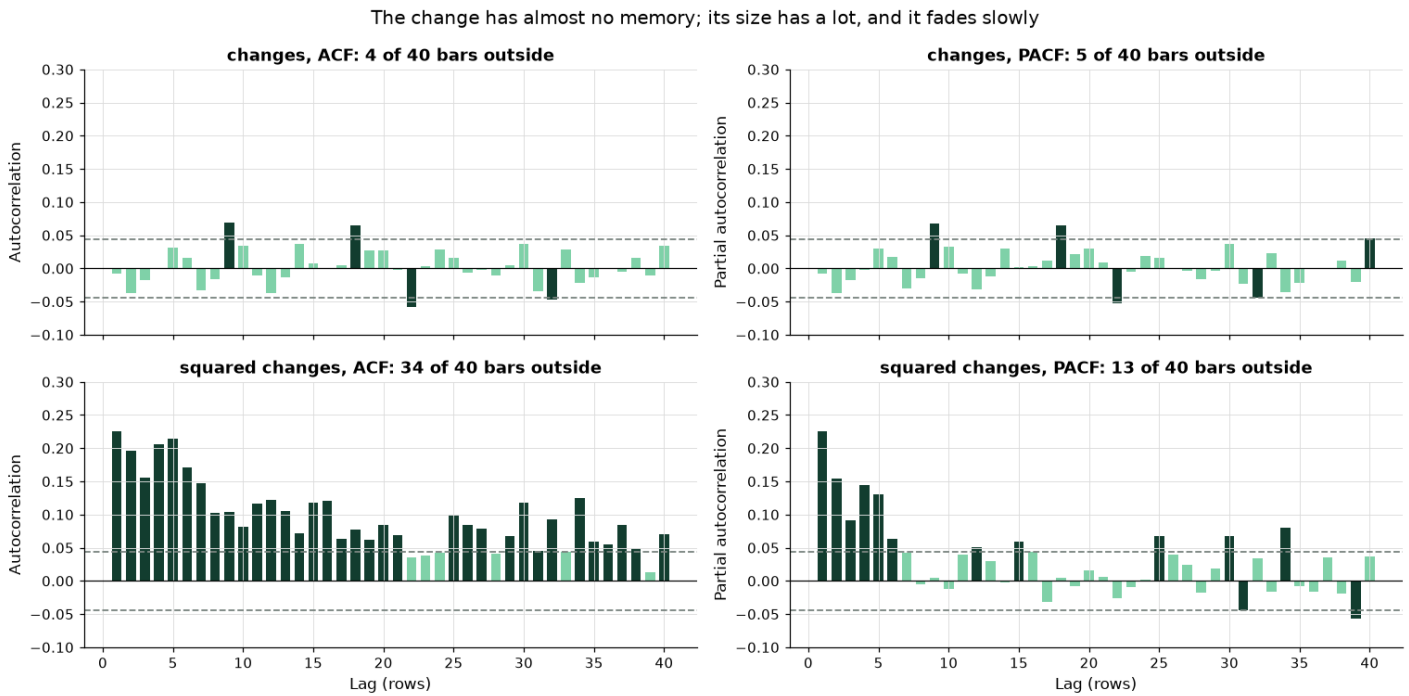
AR(p)	A weighted sum of the p values before it plus a shock: the PACF cuts off after p, the ACF decays
MA(q)	A weighted sum of the last q shocks: the ACF cuts off after q, the PACF decays
ARMA(p, q)	Both decay
Reading it	Look for the lag where one picture stops and the other keeps fading
The habit	Read the shape, not every bar: one in twenty is outside by chance

The identification grid



Bars are 2,000 simulated values; dots are the exact autocorrelations; dashed lines are the band. A bar or two beyond a cut-off is chance, not a longer model.

The change and its size



Two rows, two answers

The changes: a few bars barely outside the band, no shape; lag 1 is -0.007. The squares: 34 of 40 bars outside, the first at 0.225, fading slowly. A big move tends to follow a big move: volatility clustering, day 7.

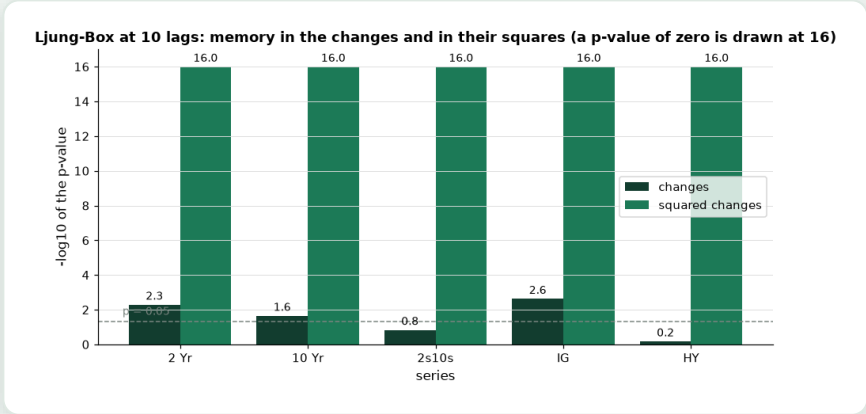
The change's memory is too small to use; the size's memory is large.

How the Ljung-Box test works

The problem	Reading bars one at a time invites the multiple-comparisons trap
The statistic	n(n + 2) times the sum over lags k of (autocorrelation at k) squared / (n - k)
Null hypothesis	No autocorrelation: Q follows a chi-squared distribution; a small p-value rejects it
You control	The lags m, and model_df (the fitted parameters) for a model's residuals
What comes out	Q and a p-value at each lag asked for

Ljung-Box on five series

Series	Changes, 10	Changes, 20	Squares, 10	Squares, 20
2 Yr	0.0051	0.0000	0.0000	0.0000
10 Yr	0.0225	0.0074	0.0000	0.0000
2s10s	0.1461	0.0252	0.0000	0.0000
IG	0.0024	0.0005	0.0000	0.0000
HY	0.6674	0.5495	0.0000	0.0000



By hand Q = 20.8096 at 10 lags for the 10 Yr (statsmodels 20.8096). The squares' p-values are zero to four decimals in every series.

Statistically detectable is not economically useful.

Excel side by side

Excel side by side (tested in Excel)

`=CORREL(D4:D2002,D3:D2001)`	CORREL -0.007480
`=SUMPRODUCT((D4:D2002-AVERAGE($D$3:$D$2002))`	by hand, lag 1: the products
`*(D3:D2001-AVERAGE($D$3:$D$2002)))/DEVSQ($D$3:$D$2002)`	divided by DEVSQ: -0.007474
`=2000*2002*SUMPRODUCT(K2:K11^2/(2000-J2:J11))`	Ljung-Box Q 20.8096
`=CHISQ.DIST.RT(M2,10)`	p-value 0.0225
`=INDEX(LINEST(D5:D2002,F5:G2002),1,1)`	PACF lag 2: -0.037348
`=1.96/SQRT(COUNT(D3:D2002))`	band 0.04383

Excel has no ACF, PACF, or Ljung-Box function: it has the pieces they are made of.

Save the day: a diff before a commit

1. Init

The work folder, local settings, the guard

2. Commits

The code, then ljung_box.csv at 10 lags

3. Change

The same file with the 20-lag columns added

4. Diff

git status, git diff --stat, git diff, then commit

```
Day 4: the Ljung-Box table at 20 lags too
Day 4: tskit with acf_by_hand
```

A diff that shows a number you did not mean to change is the signal to stop before the commit.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.

1

Docstring

Which dates fit, which dates score, what may be filled: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert what the docstring promises. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the band uses sqrt(nlags) instead of sqrt(n): for an AR(1) with phi 0.3 the function finds [].

Code that runs is not the same as code that is right.

What you can do now

- An autocorrelation by hand, equal to statsmodels'
- A bar against its band, and chance: one in twenty outside
- Every setting of acf, pacf, and acorr_ljungbox
- The identification grid: where the PACF or the ACF cuts off
- The change has almost no memory; its size has a lot

Week 1, Day 5 Case study: AR and ARIMA on the curve